

CS471 – Lesson 11 In-Class Worksheet

Non-Deterministic Search: Markov Decision Processes (MDPs)

Part 1: Guided In-Class Worksheet (Instructor-Led)

Objective: Understand non-determinism, the Markov property, and transition models.

1. Non-Determinism & Risk

Question: What does "non-deterministic" mean in the context of an autonomous UAV navigating a contested grid? How is $P(s'|s, a) = T(s, a, s')$ related to non-determinism?

Scenario: A UAV is navigating to an extraction point.

- **Case A:** The living reward is $R(s) = -0.01$ (Fuel is plentiful).
- **Case B:** The living reward is $R(s) = -0.04$ (Fuel is critically low).

Question: Identify and describe how the optimal policy changes (takes more or less risk) when the reward drops to -0.04. Why does the policy take more risk?

2. The Markov Property

The Markov Property states mathematically:

$$P(S_{t+1} = s' | S_t = s_t, A_t = a_t, S_{t-1} = s_{t-1}, A_{t-1} = a_{t-1}, \dots, S_0 = s_0)$$

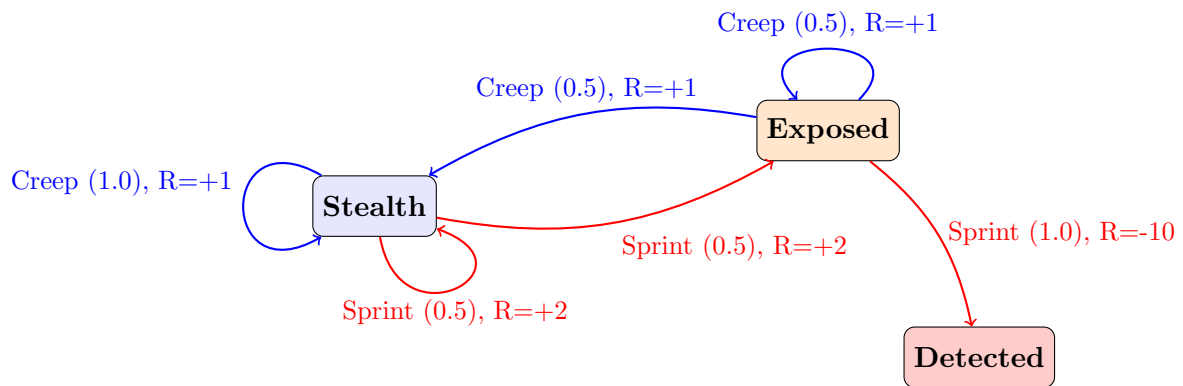
Which equals:

$$P(S_{t+1} = s' | S_t = s_t, A_t = a_t)$$

Question: State what this means in plain English. Why is this important for AI?

3. MDP Transition & Reward Table

Use the following submarine stealth Markov Network diagram to build a table of transition probabilities and rewards for each state.



S	A	S'	T(S, A, S')	R(S, A, S')
Stealth	Creep	Stealth		

Part 2: Student Independent Practice

Objective: Formulate a complete MDP from a tactical scenario and draw its network.

Electronic Warfare (EW) Operations

An Electronic Warfare (EW) aircraft operates in a contested airspace. It can be in one of three states: **Standby**, **Emitting**, or **Targeted** (which is a terminal state).

The agent has two actions available: **Loiter** and **Jam**. The environment reacts as follows:

- **In Standby:**

- Action **Loiter**: 100% chance to stay in Standby (Reward: +1).
- Action **Jam**: 50% chance to stay in Standby (Reward: +3), and a 50% chance to transition to Emitting (Reward: +3).

- **In Emitting:**

- Action **Loiter**: 50% chance to return to Standby (Reward: +1), and a 50% chance to stay in Emitting (Reward: +1).
- Action **Jam**: 100% chance to transition to Targeted (Reward: -15).

1. Build a table of transition probabilities and rewards for each state.

2. Draw the Markov Network for the problem.